4. Within the header and each record, there may be one or more fields, separated by commas. Spaces are considered part of a field and should not be ignored. The last field in the record must not be followed by a comma. For example: Red , Blue

5. Each field may or may not be enclosed in double quotes. If fields are not enclosed with double quotes, then double quotes may not appear inside the fields. For example:

```
"Red","Blue","Green"↵      #Field data with doule quotes
Black,White,Yellow         #Field data without doule quotes
```

6. Fields containing line breaks (CRLF), double quotes, and commas should be enclosed in double-quotes. For example:

```
Red, ",", Blue CRLF    # comma itself is a field value.so it is enclosed with double  quotes
Red, Blue , Green
```

7. If double-quotes are used to enclose fields, then a double-quote appearing inside a field must be preceded with another double quote. For example:

```
""Red"", ""Blue"", ""Green"" CRLF    # since double quotes is a field value it is enclosed with another
, ,  White                            double  quotes
```

Note

The last row in the above example (, , White) begins with two commas because the first two fields of that row were empty in our spreadsheet. Don't delete them — the two commas are required so that the fields correspond from row to row. They cannot be omitted.

13.5 Create A CSV File Using Microsoft Excel

To create a CSV file using Microsoft Excel, launch Excel and then open the file you want to save in CSV format. For example, below is the data contained in our sample Excel worksheet:

12th Computer Science_EM Chapter 13.indd 230 23-12-2022 15:39:37